B.Tech. Examination-2024-2025

Electronics and Communication Engineering Department

(Odd Semester Regular and Supplementary)

Course Title - Basic Electronics Engineering Course Code - ECEUGES01

Full Marks : 80 **Time-3 hrs**

(Answer any five, Symbols have their usual meaning)

1. (a) Compare between metal, semiconductor and insulator concerning their band diagram. 3
   (b) Explain electron-hole pair generation in intrinsic Si semiconductor at room temperature. Can it behave as an insulator at room temperature? Explain 3+2
   (c) How n-type semiconductor is formed ? 4
   (d) Draw the energy band diagram of intrinsic Si-semiconductor. 4
2. (a) Write the Fermi-Dirac distribution function. 3
   (b) Show that Fermi function is symmetrical about E_F i.e. $f(E_F + \Delta E) = 1 - f(E_F - \Delta E)$. 8
   (c) Explain why in n-type semiconductor the Fermi level energy E_F is close to the ~~valence~~ band edge E_v of the band diagram. 5
   conduction
3. (a) Show that equilibrium concentrations of electrons (n_0) in a semiconductor can be expressed as :- 3+3

$$n_0 = N_c e^{-\left(\frac{E_c - E_F}{KT}\right)}$$

   (b) Show that intrinsic carrier concentration under equilibrium is constant and can be expressed as :- $n_i = \sqrt{N_C N_V} e^{-\frac{E_g}{2KT}}$. 6
   (c) An abrupt Si p-n junction has $N_a = 10^{18}\ cm^{-3}$ on one side and $N_d = 5 \times 10^{15}\ cm^{-3}$ on the other. Calculate the contact potential V_0. 4
4. (a) Write down Shockley diode equation for p-n junction diode. 3
   (b) Show that diode dynamic resistance in forward bias is $r_d = \frac{26mV}{I_D}$. 5
   (c) Show that cut-in voltage of p-n junction diode is expressed as : 8

$$V_\gamma = V - \eta V_T \left(1 - e^{-\frac{V}{\eta V_T}}\right)$$

5. (a) Explain the formation of depletion region and barrier potential V_0 ina p-n junction diode under equilibrium. 8
   (b) Explain the avalanche and zener breakdown mechanism in a p-n junction diode. 6
   (c) Draw the ideal diode current-voltage characteristics. 2
6. (a) Draw and explain the output waveform of the circuit in Fig. 1. 5
   (b) Draw and explain the output waveform of the circuit shown in Fig. 2. 5
   (c) Show that in a full wave rectifier circuit output only even cosine harmonics are present. 6

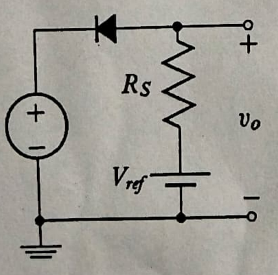

Fig. 1

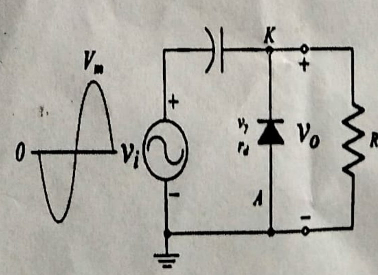

Fig 2

7. (a) Establish the relation for rectifier output, $V_{r(rms)} = \sqrt{V_{rms}^2 - V_{dc}^2}$. 5
   (b) Show that the output DC voltage (V_{dc}) and ripple factor of full wave rectifier output is : (i) $V_{dc} = \frac{2V_m}{\pi}$ (ii) $\Gamma = 48.3\%$ 6
   (c) Draw and explain the output waveform of the circuit in Fig. 3. 5

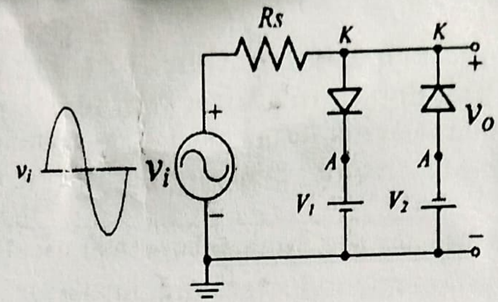

Fig. 3

8. (a) Explain different current components in a p-n-p transistor in active region with a suitable diagram. 6
   (b) Establish the relation between current amplification factors α and β. 4
   (c) Briefly explain the CB, CE and CC mode of operation in a transistor. 6

9. (a) Determine V_L, I_L, I_Z and I_R for the network of Fig. 1 if $R_L = 180\ \Omega$, $R_s = 220\ \Omega$ and $P_{zmax} = 400$ mW. 12
   (b) Repeat part (a) if $R_L = 330\ \Omega$.
   (c) Determine the value of R_L that will establish maximum power conditions for the Zener diode.
   (d) Determine the minimum value of R_L to ensure that the Zener diode is in the "on "state.

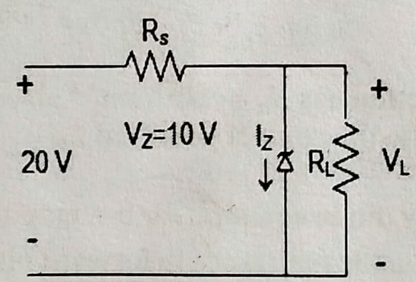

Fig. 4

   (e) Define line regulation and load regulation for the zener diode to be used as a Voltage regulator. 4

ALIAH UNIVERSITY

Name of the Examination: Even Semester **Year: 2024-25**
Subject: Engineering Chemistry **Course code: CHMUGBS01**
Duration: 3 Hrs **Full Marks: 80**

Use separate answer scripts for each group

Group A

01. Explain the following terms (answer any FOUR): (a) Path function (b) Irreversible process (c) Molar heat capacity (c) Enthalpy (d) Extensive property (e) Joule Thomson phenomenon — 2.5×4

answer any FOUR from the following questions

02. (a) Derive the expression for the work done during the isothermal, reversible expansion of an ideal gas.(b) Draw the pressure-volume (P-V) diagrams depicting the isothermal expansion of an ideal gas, clearly differentiating between reversible and irreversible processes. — 2.5+2.5
03. (a) State the thermodynamic condition for equilibrium. (b) Show Joule Thomson effect is an isoenthalpic process. — 2.5+2.5
04. (a) Prove mathematically that work done in a reversible process is more than that of an irreversible process (b) Differentiate between reversible and irreversible processes. — 2.5+2.5
05. a) Explain the concepts of C_p and C_v for gases. — 2.5+2.5
    (b) Why is C_p always greater than C_v? Establish the relationship between them.
06. (a) Find the work done when one mole of the gas is expanded reversibly and isothermally from 5 atm to 1 atm at 25^0C.(b) 1 gm of water is converted into steam at the same temperature. The volume of water becomes 1671 ml ob boiling. Calculate the change in the internal energy of the system if the heat of vapourization is 540 cal/gm. — 2.5+2.5

Group B (Answer any TEN) — 5× 10=50

07. Give brief account of
    i) Polythene ii) Compounding of rubber
08. (a)Why are silicones called inorganic polymers? (b) Discuss the synthesis procedure of linear chain silicones.
09. (a) What is meant by green chemistry?
    (b) Write informative notes on the mechanism of radical polymerisation
10. Write notes on: (a) Polystyrene (PS) (b) Polyvinyl Chloride (PVC).
11. What is meant by the vulcanisation of rubber? How vulcanisation takes place? Write its uses?
12. What is Biopolymers? How is it prepared? Give some importance of Bio-ploymers.
13. Write down the importance of green chemistry. Give the preparation of Aspirin from salicylic acid using microven technique. Why this technique is preferred.
14. Explain with examples: i) Gutta percha ii) Conducting polyaniline
15. a) What is Ziegler Natta catalyst with some application. b) How will you prepare Styrene-Butadiene Rubber (SBR). Mention its uses.
16. a) What are drawbacks of natural rubber? (b) How will you prepare Novalac resin from phenol and formaldehyde with mechanism
17. a) What is meant by degree of polymerisation? b) Distinguish between addition and condensation polymerization.
18. How the following are prepared? Mention their uses (a) Teflon (b) LDPE
19. a) How is the Crepe rubber obtained from latex? b) What is meant by degree of polymerisation?
20. a) Distinguish between addition and condensation polymerisation with examples. b) Give a short note on microwave radiation in green synthesis.

End-Semester Examination - 2025 (Spring)
MATUGBS02
(Engineering Mathematics II)
Semester – II (4 Year B.Tech)

Full Marks – 80 **Time – 3 Hours**

Notations and symbols have their usual meaning.
Use separate answer script for Group-A and Group-B

GROUP-A

Section A. Answer ALL questions. Each question carries TWO mark.

1. (a) Define skew symmetric matrix with example.

   (b) Show that any square matrix can be written as sum of two matrices, one is symmetric and another is skew symmetric.

   (c) Obtain the determinant of the matrix

$$A = \begin{bmatrix} 0 & a & -b \\ -a & 0 & c \\ b & -c & 0 \end{bmatrix}.$$

   (d) Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}, B = \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix},$$

   show that $AB = BA$.

   (e) Define rank of a matrix.

Section B. Answer any THREE questions. Each question carries 10 marks.

2. (a) Define echelon form of a matrix with example. Reduce the echelon form of matrix A, where

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 4 & -1 & 5 \\ 2 & 0 & 6 \end{bmatrix},$$

   Hence comment on the rank of A 3+6+1

3. (a) Show that $Det(A) = (a-b)(a-c)(b-c)$, where

$$A = \begin{bmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{bmatrix}.$$

4

(b) Find two matrices P and Q such that $PAQ = I_3$, where

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 3 & 0 \\ 6 & 2 & 3 \end{bmatrix},$$

6

4. (a) Let $D_n = Det((a_{ij}))_{n,n}$, where $a_{ij} = a^{|i\ j|}$, $a > 0$, show that $D_n = (1-a^2)D_{n-1}$. 4

(b) Solve the following system of equations using Cramer's rule:

$$\begin{aligned} x - 2y + 2z - 2 &= 0 \\ 2x - y - 2z - 1 &= 0 \\ 2x + 2y + z - 7 &= 0. \end{aligned}$$

6

5. (a) If A is a non-singular matrix then show that (i) A^{-1} (ii) $Adj(A)$ is also non-singular. Find the inverse of the matrix A, where

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & 4 \\ 3 & 3 & 7 \end{bmatrix};$$

(2+2)+6

6. (a) Define orthogonal matrix. Check whether the below matrix is orthogonal or not.

$$A = \begin{bmatrix} 2/3 & 1/3 & 2/3 \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/3\sqrt{2} & -4/3\sqrt{2} & 1/3\sqrt{2} \end{bmatrix},$$

2+8

GROUP - B (40 marks)

Answer all the following questions:

(a) The solution of the given exact differential equation $(\frac{1}{y} - \frac{2}{x})dx - (\frac{x}{y^2} - \frac{3}{y})dy = 0$ is

i. $\log(\frac{y^3}{x^2}) + \frac{x}{y} = c$

ii. $\log(\frac{y^2}{x^2}) + \frac{x}{y} = c$

iii. $\log(\frac{y^3}{x^3}) + \frac{x}{y} = c$

iv. $\log(\frac{y}{x}) + \frac{x}{y} = c$

(where c is arbitrary constant) [2]

(b) Which of the following options is integrating factor of the given differential equation $y(2xy + e^x)dx - e^x dy = 0$

i. $\frac{1}{x^2}$

ii. $\frac{1}{xy}$

iii. $\frac{1}{y^2}$ [2]

iv. $\frac{1}{x^2y^2}$

(c) Which of the following options is the particular integral of the given differential equation $(D^2-1)y = 2e^x$ where $D \equiv \frac{d}{dx}$

i. xe^x

ii. x

iii. e^x

iv. e^{x^2} [2]

(d) Find mod z and amp z (principal amplitude) where $z = \frac{1+\sqrt{3}i}{1+i}$

i. $\sqrt{2}, \frac{\pi}{2}$

ii. $\sqrt{2}, \frac{\pi}{4}$

iii. $\sqrt{3}, \frac{\pi}{2}$

iv. $\sqrt{3}, -\frac{\pi}{4}$ [2]

(e) If $z_1 = 1+i$ and $z_2 = 1+\sqrt{3}i$, then the principle amplitude of z_1z_2 is

i. $\frac{7\pi}{12}$

ii. $\frac{9\pi}{12}$

iii. $\frac{3\pi}{4}$

iv. $\frac{2\pi}{3}$ [2]

(f) **Answer any SIX of the following questions:** $(6\times5=30)$

i. Solve the linear differential equation $\frac{dy}{dx} + \frac{3x^2}{1+x^3}y = \frac{1}{x^4}$. [5]

ii. Test for exactness and solve [5]
$x^2(1+x)D^2y + 2x(2+3x)Dy + 2(1+3x)y = 0.$ [5]

iii. Solve $(x^3D^3 - x^2D^2 + 2xD - 2)y = x^3 + 3x$. [5]

iv. Solve the Bernoulli's differential equation $\frac{dy}{dx} - 3x^2y = -e^{x}\ 2x^3\ y^3$. [5]

v. Solve the differential equation by using the method 'Variation of Parameter'

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = \frac{e^{x}}{x^2}.$$

[5]

vi. Solve: $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + \frac{dy}{dx} = 3x+1$. [5]

vii. Solve: $\frac{d^2y}{dx^2} - 4y = xe^{2x}$. [5]

viii. Solve: $\frac{d^2y}{dx^2} + a^2y = \sin ax$.

ix. Show that the solution of the differential equation (Clairaut's form) $y = px + f(p)$ is $y = cx + f(c)$ where c is an arbitrary constant and hence solve $y = px + p^2 + 2p + 1$. [5]

x. (i) If one of the root of the equation $x^3 - 10x^2 + ax - b$ (where $a, b \in \mathbb{R}$) is $4+i$, then find other two roots and also find the value of a and b. [5]

OR, (ii) If x, y, z are positive real numbers and $x+y+z=1$, prove that [5]

$$8xyz \leq (1-x)(1-y)(1-z) \leq \frac{8}{27}$$

Aliah University

Dept. of Computer Science & Engineering

End-Semester (Spring Semester) **Examination-2025**

(Question Paper for **2nd Semester** **Section- A**)

Paper Name: Programming for Problem Solving

Full Marks: 80

Paper Code: CSEUGES01

Times: 3 Hours

GROUP-A

(Multiple Choice Type Question)

1) Answer any ten Questions (choose the right option):- (1X10=10)

a) Array is the following type of memory allocation: [CO3, BTL1]
(i) continuous and homogeneous, (ii) continuous and heterogeneous, (iii) discrete and homogeneous (iv) discrete and heterogeneous.

b) The calloc() function associated with the following type of memory allocation: [CO4, BTL1]
(i) static, (ii) dynamic, (iii) both of these, (iv) none of these.

c) In 'C', '\0' stands for (i) zero, (ii) null character, (iii) string, (iv) End of File. [CO3, BTL1]

d) 'structure' has the following type of elements: [CO5, BTL1]
(i) homogeneous, (ii) heterogeneous, (iii) both of these, (iv) none of these.

e) 'EOF' is related to: [CO6, BTL1]
(i) string, (ii) FILE, (iii) array, (iv) structure.

f) The string size and the array size in the initializer- [CO5, BTL4]
char team[100] = "soldier";
are:
(i) 7 and 7, (ii) 100 and 100, (iii) 7 and 100, (iv) 100 and 7, respectively.

g) Mode 'a' in FILE concerns with: (i) appending data, (ii) reading data, (iii) writing data, (iv) addressing data. [CO6, BTL1]

h) Find the correct: (i) int *p, i=100; p=i; (ii) int *p, i=100; p=&i; (iii) int *p; float j=100.123426; p= &j; (iv) int p*, i=100; p=&i; [CO5, BTL4]

i) The function ***rewind(fp)*** takes the file pointer ***fp*** and resets the position to (i) the middle of the FILE, (ii) the start of the FILE, (iii) the end of the FILE, (iv) the current position of the FILE. [CO6, BTL1]

j) 'sizeof()' is a / an (i) default function, (ii) user defined function, (iii) library function, (iv) operator [CO5, BTL1]

k) Int i=10; i= 5>10; value of variable i for the code segment is: (i) 10, (ii) 0, (iii) 1, (iv) 5. [CO1, BTL5]

GROUP-B

(Answer any five) **(5X5=25)**

2) What is increment operator? Distinguish pre and post increment with suitable example. [CO1, BTL4]

3) When to use loops? Write small program to explain the function of break and continue. [CO1, BTL1 & BTL5]

4) Conditional operator is equivalent to if else statement. Justify. Write a program to check whether a given number is even or not using tertiary operator. [CO1, BTL5]

5)
```
#include<stdio.h>
int main(){
int i;
while(i)
{
printf("%4d",i)
}
return 0;
}
```
What will be the output of the above code (a) if the variable i is initialized with 0 and (b) if the variable i is initialized with 1. Justify your answer. [CO3, BTL6]

6)
```
#include<stdio.h>
int main(){
```

Department of English
End Semester Examination, 2025
(For the 2nd Semester Students of all the Departments of the Engineering Faculty)
ENGUGHU01: Communicative English

Full Marks: 80

Time: 3 Hours

Section I

1. Answer **ANY TEN** of the following questions: **2x10=20Marks**

1. What are the channels of communication?
2. What is the basic difference between general communication and professional communication?
3. What is intra personal communication?
4. Exemplify a barrier in effective listening.
5. Give an example of Grapevine communication.
6. What is code?
7. How many diphthongs are there in English language?
8. What is 'LSRW'?
9. What is phoneme?
10. What is *lingua franca*?
11. What is semantic gap?
12. What is skimming in reading?
13. What is homophone? Give an example.
14. What is phrase? Give an example.
15. What is monologue?
16. Give two examples of idioms in English language.

Section II

2. Answer **ANY FOUR** of the following questions: (word limit 200) **5x4=20 Marks**

1. What are the characteristics of a good communication?
2. What is business communication? What are the methods of business communication?

3. Discuss the barriers of effective listening?
4. Write a short dialogue between two friends discussing the price hike.
5. What is the difference between summary and paraphrase?
6. Write a short report on the metro rail in Kolkata.
7. What are the important features of close reading?

Section III

3. Answer **ANY FOUR** of the following questions. (Word limit 400 **10X4=40 Marks**

1. Define Communication. What are the functions of Communication? (5+5)
2. Define and discuss the differences between Verbal Communication and Non-verbal Communication. (10)
3. Define Grapevine Communication. What are the demerits of Grapevine Communication? (5+5)
4. Explain the different flows of communication in the business organisations.(10)
5. Attaching a Resume, write a job application for the post of a librarian in a university. (10)
6. Mention the different Presentation strategies of Speaking Skills. (10)
7. Write a report of the Annual Teachers Day celebration in your university. (10)